

All 268 competencies, week by week.

Human Physiology · Section BIOL-5-D9286 · Sutter Internet (NET)

15 weeks · September 8 to December 16, 2026 · three midterms

What this is. Every single thing this course will ask you to be able to do, sorted into the week it is taught. If it is not on this list, it is not on an exam. If it is on this list, it is fair game on the midterm that covers its week.

How to use it. Find your week. Those are your competencies. Your note sheet already has one page for each of them, so you do not have to copy anything off this list. Come back to it before a midterm and read down the five weeks it covers. Anything you cannot do without looking is your study list.

A note on the other packet. The Competency Packet in Canvas holds these same 268 grouped by unit, with the entry expectations in front. That one is for transfer and equivalency review. This one is for finding your week.

Dr. Sharilyn Rennie

Weeks 1 to 5

71 competencies. Midterm 1. Window Monday October 12 at 8 am to Sunday October 18 at 10 pm.

Week 1 How physiology works and what keeps you steady

Tue Sep 8 to Sun Sep 13 · 12 competencies

1 Physiology and levels of function

Define physiology and place a given process at the correct level of organization from molecule to organism, then state the level at which that process is best explained.

2 Structure and function relationship LAB

Predict how a change in the structure of a molecule, cell, tissue, or organ alters its function, using examples from two different organ systems.

3 Homeostasis defined

Define homeostasis, regulated variable, and setpoint, and distinguish homeostasis from chemical equilibrium and from steady state.

4 Feedback loop components LAB

Diagram a negative feedback loop labeling stimulus, sensor, afferent path, integrating center, efferent path, effector, and response, and trace body temperature or blood glucose through every step.

5 Negative and positive feedback

Distinguish negative from positive feedback by the direction of the response and give a physiological example of each, explaining why a positive feedback loop needs an outside event to end it.

6 Feedforward control and acclimatization

Explain anticipatory feedforward control and acclimatization and identify which one is operating in a given scenario.

7 Local and reflex control pathways

Distinguish a local control pathway from a long distance reflex pathway and classify a given response as one or the other.

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- 8 Mass balance** LAB
Apply the mass balance equation to a solute or to body water and calculate the intake, production, and output combination that holds the amount in the body constant.
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- 9 Units and unit conversion** LAB
Convert among the units used in physiology including molarity, osmolarity, milliequivalents, mmHg, liters per minute, and percent solutions.
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- 10 Graphing and data interpretation** LAB
Construct a labeled graph with the independent variable on the x axis, and read slope, direction, and trend from a physiological data set.
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- 11 Experimental design and controls** LAB
Identify the hypothesis, independent variable, dependent variable, and control condition in a physiology experiment and state what the control rules out.
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- 12 Measurement error and variability** LAB
Distinguish random from systematic error and explain why physiological measurements are repeated and averaged.
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Week 2 The chemistry that does work in the body

Mon Sep 14 to Sun Sep 20 · 7 competencies

- 13 Water and solution properties**
Explain how the polarity and hydrogen bonding of water determine solubility and identify whether a given solute is hydrophilic or hydrophobic.
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- 14 pH and buffers**
Define pH, state the normal pH range of arterial blood, and explain how a buffer pair resists a change in pH when acid or base is added.
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- 15 Protein structure and function**
Relate the levels of protein structure to binding site shape and explain how denaturation by heat or pH change destroys function.
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- 16 Enzyme activity and regulation**
Describe how enzymes lower activation energy and predict the effect of substrate concentration, temperature, pH, and competitive or allosteric inhibition on reaction rate.
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17 ATP and energy coupling

Explain how ATP hydrolysis is coupled to endergonic cellular work and name three categories of work that require it.

18 Enzyme assay

Measure enzyme activity across a range of temperature or pH using a spectrophotometric or colorimetric assay, plot the results, and identify the optimum.

19 ATP production pathways

Compare glycolysis, the citric acid cycle, and oxidative phosphorylation by location, oxygen requirement, and ATP yield.

Week 3 Getting across the membrane

Mon Sep 21 to Sun Sep 27 · 16 competencies

20 Body fluid compartments

State the approximate volumes of total body water, intracellular fluid, extracellular fluid, plasma, and interstitial fluid in a 70 kg adult and compare the dominant solutes of the intracellular and extracellular compartments.

21 Compartment separation and clinical volume shifts

Predict the direction of water movement between compartments when extracellular osmolarity rises or falls and name a clinical situation that produces each shift.

22 Membrane composition and fluidity

Describe the fluid mosaic membrane and state how phospholipids, cholesterol, glycolipids, and integral and peripheral proteins contribute to its properties.

23 Determinants of permeability

Rank molecules by their ability to cross a lipid bilayer unaided using size, charge, and lipid solubility, and predict which will require a protein.

24 Simple diffusion and Fick's law

State the variables in Fick's law of diffusion and predict how a change in concentration gradient, surface area, membrane thickness, or distance changes the rate of transfer.

25 Osmolarity and tonicity

Calculate the osmolarity of a solution, distinguish osmolarity from tonicity, and classify a solution as isotonic, hypotonic, or hypertonic to a cell.

26 Osmosis and cell volume

Predict the direction of water movement and the resulting change in cell volume when a cell is placed in a solution of stated osmolarity and penetrating solute content.

27 Diffusion and osmosis experiment

Measure diffusion and osmotic movement across a selectively permeable membrane and relate the observed rate to molecular size and concentration gradient.

28 Tonicity and red blood cells

Observe erythrocytes in solutions of different tonicity, identify crenation, normal shape, and hemolysis, and explain each result.

29 Facilitated diffusion

Describe carrier and channel mediated diffusion and explain why carrier mediated transport shows saturation and specificity while simple diffusion does not.

30 Primary active transport

Explain how the sodium potassium ATPase uses ATP to move three sodium out and two potassium in, and state the two gradients it maintains.

31 Secondary active transport

Distinguish symport from antiport and trace how the sodium gradient powers glucose uptake by SGLT and calcium removal by the sodium calcium exchanger.

32 Transport maximum and saturation

Interpret a transport rate curve, identify the transport maximum, and apply the concept to renal glucose handling in hyperglycemia.

33 Vesicular transport

Compare phagocytosis, pinocytosis, receptor mediated endocytosis, and exocytosis by trigger, cargo, and energy requirement.

34 Transepithelial transport

Trace glucose or sodium from lumen to blood across a polarized epithelium and identify which step occurs at the apical membrane and which at the basolateral membrane.

35 Transport simulation

Use a membrane transport simulation to distinguish simple diffusion from facilitated diffusion and active transport by their response to gradient reversal and metabolic poison.

Week 4 How cells talk, and the electrical signal

Mon Sep 28 to Sun Oct 4 · 20 competencies

36 Ion distribution and electrochemical gradients

State the typical intracellular and extracellular concentrations of sodium, potassium, chloride, and calcium and separate the chemical from the electrical component of the driving force on each ion.

37 Nernst equation

Calculate the equilibrium potential for an ion with the Nernst equation and explain what the sign of the result means for the direction that ion will move.

38 Resting membrane potential

Explain why the resting membrane potential sits near the potassium equilibrium potential and predict how it shifts when membrane permeability to potassium or sodium changes.

39 Ion channel gating

Compare leak, voltage gated, ligand gated, and mechanically gated channels by what opens each one and give a physiological location for each.

40 Depolarization and hyperpolarization

Define depolarization, repolarization, hyperpolarization, and overshoot and label each on a membrane potential tracing.

41 Membrane potential simulation

Manipulate extracellular potassium and sodium in a simulation and record the resulting change in resting membrane potential against the Nernst prediction.

42 Neuron structural and functional classes

Classify a neuron as multipolar, bipolar, or pseudounipolar by structure and as sensory, motor, or interneuron by function, and match each class to a location in the nervous system.

43 Functional regions of a neuron

Label the dendrites, cell body, axon hillock, trigger zone, axon, and axon terminal and state which signal type each region carries.

44 Glial cell functions

Match astrocytes, oligodendrocytes, microglia, ependymal cells, satellite cells, and Schwann cells to their physiological roles.

45 Myelin and its loss

Explain how myelin and the nodes of Ranvier speed conduction and predict the functional consequence of demyelination.

46 Axonal transport

Distinguish fast anterograde and retrograde axonal transport from slow transport and state what each carries.

47 Graded potentials

Describe how a graded potential is produced and explain why it varies with stimulus strength and decays with distance, using the terms current leak and cytoplasmic resistance.

48 Action potential phases

Diagram an action potential and state the channel state and ion movement responsible for depolarization to peak and for repolarization and afterhyperpolarization.

49 Threshold and all or none

Explain what threshold represents at the trigger zone and why action potential amplitude does not change with a stronger stimulus.

50 Coding of stimulus intensity

Explain how the nervous system encodes stimulus strength when action potential amplitude is fixed.

51 Refractory periods

Distinguish the absolute from the relative refractory period by channel state and explain how the refractory period sets maximum firing frequency and prevents backward conduction.

52 Conduction velocity

Rank axons by conduction velocity using diameter and myelination and contrast continuous with saltatory conduction.

53 Effects of altered extracellular ions

Predict the effect of hyperkalemia and hypokalemia on resting potential and excitability and explain why local anesthetics that block sodium channels abolish the action potential.

54 Action potential simulation

Run a neuron simulation with sodium and potassium channel blockers and interpret the resulting change in the action potential tracing.

55 Nerve conduction measurement

Measure conduction velocity from a recorded compound action potential and account for the difference between the fastest and slowest fibers.

Week 5 Synapses and central integration

Mon Oct 5 to Sun Oct 11 · 16 competencies

56 Sequence at a chemical synapse

Order the events of chemical synaptic transmission from action potential arrival through calcium entry and vesicle fusion to receptor binding and postsynaptic response.

57 Neurotransmitter classes

Match acetylcholine, the catecholamines, serotonin, glutamate, GABA, glycine, and the neuropeptides to their usual excitatory or inhibitory effect and to a site of action.

58 Neurotransmitter removal

Name the three routes that clear a neurotransmitter from the synaptic cleft, degradation, reuptake, and diffusion, and give a drug that blocks one of them.

59 Excitatory and inhibitory postsynaptic potentials

Distinguish an EPSP from an IPSP by the ion channel opened and the direction of the membrane potential change.

60 Summation and integration

Distinguish temporal from spatial summation and determine whether a stated combination of EPSPs and IPSPs will bring the trigger zone to threshold.

61 Presynaptic modulation

Compare presynaptic inhibition and facilitation with postsynaptic modulation and state the advantage of presynaptic control at a single input, which is selectivity.

62 Synaptic plasticity

Explain long term potentiation as a mechanism of learning and identify the roles of repeated stimulation and receptor insertion.

63 Electrical synapses

Compare an electrical synapse at a gap junction with a chemical synapse by speed and by capacity for modulation.

64 Synapse simulation

Alter neurotransmitter release and receptor availability in a synapse simulation and interpret the resulting postsynaptic recording.

65 Reflex arc components

Diagram a reflex arc naming the receptor, sensory neuron, integrating center, motor neuron, and effector, and classify the reflex as monosynaptic or polysynaptic.

66 Muscle spindle and stretch reflex

Explain how the muscle spindle detects stretch and trace the stretch reflex including reciprocal inhibition of the antagonist.

67 Golgi tendon organ

Contrast the stimulus and the reflex response of the Golgi tendon organ with those of the muscle spindle.

68 Withdrawal and crossed extensor reflexes

Trace the flexor withdrawal reflex and the crossed extensor reflex and explain how the two together preserve balance.

69 Ascending and descending pathways

Compare the dorsal column, spinothalamic, and corticospinal pathways by the information carried, the side of decussation, and the deficit produced by a lesion.

70 Cerebrospinal fluid and the blood brain barrier

State the functions of cerebrospinal fluid and explain how the blood brain barrier selects what reaches neurons, including why lipid soluble drugs cross readily.

71 Reflex testing and reaction time

Elicit and grade deep tendon reflexes and measure reaction time to distinguish a reflex response from a voluntary response by latency.

Weeks 6 to 10

103 competencies. Midterm 2. Window Monday November 16 at 8 am to Sunday November 22 at 10 pm.

Week 6 Sensing the world, and the responses you do not control

Mon Oct 12 to Sun Oct 18 · 24 competencies

72 Sensory transduction

Explain how a stimulus is converted into a receptor potential and how that graded potential becomes a train of action potentials in the sensory neuron.

73 Receptor classification

Classify a receptor by stimulus type as a mechanoreceptor, chemoreceptor, thermoreceptor, photoreceptor, or nociceptor and give a body location for each.

74 Stimulus coding

Explain how the nervous system encodes modality, location, intensity, and duration and identify the labeled line principle.

75 Receptive fields and acuity

Relate receptive field size and lateral inhibition to two point discrimination and predict which body regions have the finest acuity.

76 Receptor adaptation

Distinguish tonic from phasic receptors by their adaptation rate and match each to a stimulus the body needs to monitor continuously or only at onset.

77 Somatosensory pathways

Trace touch and pain information from receptor to cortex and explain why the somatosensory homunculus is disproportionate.

78 Pain and its modulation

Distinguish fast and slow pain by fiber type, explain referred pain by convergence, and describe how gate control and endogenous opioids reduce pain transmission.

79 Tactile mapping and adaptation testing

Measure two point discrimination at several body sites and show receptor adaptation and referred sensation experimentally.

80 Optics and accommodation

Explain refraction by the cornea and lens and describe how accommodation and the pupillary reflex focus near and far images.

81 Phototransduction

Trace phototransduction from photon absorption by retinal through the change in cyclic GMP to the decrease in glutamate release and explain why photoreceptors hyperpolarize to light.

82 Rods cones and visual processing

Compare rods and cones by sensitivity and acuity and trace the visual pathway to predict the field deficit produced by a lesion at the optic nerve or the optic chiasm.

83 Refractive errors and clinical vision

Explain myopia, hyperopia, presbyopia, and astigmatism in terms of focal point and eyeball or lens shape and state the corrective lens for each.

84 Auditory transduction

Trace sound from the tympanic membrane through the ossicles and cochlear fluid to hair cell bending and auditory nerve firing and explain how pitch and loudness are coded.

85 Conductive and sensorineural hearing loss

Distinguish conductive from sensorineural hearing loss by the site of the lesion and by the expected Weber and Rinne results.

86 Equilibrium

Explain how the semicircular canals detect angular acceleration and how the utricle and saccule detect linear acceleration and head position.

87 Taste and smell

Compare gustatory and olfactory transduction and explain why olfactory input reaches the limbic system without a thalamic relay.

88 Vision testing

Perform visual acuity, blind spot, accommodation, and color vision testing and interpret each result.

89 Hearing and equilibrium testing

Perform Weber and Rinne tuning fork tests and a balance test and interpret the findings.

90 Autonomic and somatic organization

Contrast the somatic motor system with the autonomic two neuron pathway by neuron count and by effector tissue.

91 Sympathetic and parasympathetic divisions

Compare the two autonomic divisions by CNS origin and ganglion location and predict the effect of each on heart rate and on the gastrointestinal tract.

92 Autonomic neurotransmitters and receptors

Match nicotinic, muscarinic, alpha, and beta receptors to their neurotransmitter, their location, and the response each produces.

93 Dual innervation and tonic control

Explain dual innervation and autonomic tone and predict the effect of removing one division from a dually innervated organ.

94 Autonomic pharmacology

Predict the effect of a beta blocker or a muscarinic antagonist on a named organ from the receptor it targets.

95 Autonomic function testing

Measure heart rate variability and the response to a cold pressor or orthostatic challenge and interpret the result as sympathetic or parasympathetic dominance.

Week 7 Muscle, and how movement gets commanded

Mon Oct 19 to Sun Oct 25 · 24 competencies

96 Neuromuscular junction

Trace transmission at the neuromuscular junction from the motor neuron action potential to the end plate potential and explain why every action potential in the motor neuron produces one in the muscle fiber.

97 Excitation contraction coupling

Trace excitation contraction coupling from the T tubule through the DHP and ryanodine receptors to calcium release from the sarcoplasmic reticulum.

98 Crossbridge cycle

Order the steps of the crossbridge cycle, binding, power stroke, detachment, and reactivation, and state where ATP binds and where it is hydrolyzed.

99 Calcium and the regulatory proteins

Explain how calcium binding to troponin moves tropomyosin off the myosin binding site and predict what happens to contraction if calcium cannot be released.

100 Relaxation and calcium removal

Explain how SERCA and acetylcholinesterase end a contraction and predict the result if either one fails.

101 The muscle twitch

Label the latent period, contraction phase, and relaxation phase on a twitch tracing and explain what limits each phase.

102 Summation and tetanus

Explain wave summation and distinguish unfused from fused tetanus by stimulus frequency and calcium accumulation.

103 Motor units and recruitment

Define the motor unit and explain how size of the unit and orderly recruitment grade the force a whole muscle produces.

104 Length tension relationship

Interpret a length tension curve and explain why tension falls at very short and very long sarcomere lengths.

105 Isometric and isotonic contraction

Distinguish isometric, concentric, and eccentric contractions and predict shortening velocity from the load applied.

106 Muscle fiber types

Compare slow oxidative, fast oxidative glycolytic, and fast glycolytic fibers by speed, ATP source, fatigue resistance, and typical task.

107 ATP sources during activity

Order creatine phosphate, anaerobic glycolysis, and oxidative phosphorylation by how quickly each supplies ATP and how long each can sustain it.

108 Fatigue and oxygen debt

Distinguish central from peripheral fatigue and explain the metabolic basis of excess post exercise oxygen consumption.

109 Muscle plasticity

Predict the change in fiber size, capillary supply, and mitochondrial content produced by endurance training, resistance training, and disuse.

110 Electromyography and grip fatigue

Record grip force and surface EMG over a sustained contraction and relate the decline in force to recruitment and fatigue.

111 Twitch and tetanus simulation

Generate twitch and tetanus tracings in a muscle simulation by varying stimulus voltage and frequency and identify threshold and maximal stimulus.

112 Smooth muscle contraction mechanism

Trace smooth muscle contraction through calcium entry and release to calmodulin and myosin light chain kinase and explain why the mechanism is slower and more economical than in skeletal muscle.

113 Smooth muscle types and regulation

Compare single unit and multi unit smooth muscle and list the stimuli that alter smooth muscle tone including stretch and local chemical signals.

114 Cardiac muscle properties

Explain how intercalated discs and gap junctions allow the myocardium to act as a functional syncytium and state why cardiac muscle cannot be tetanized.

115 Comparison of the three muscle types

Build a comparison of skeletal, cardiac, and smooth muscle by calcium source, regulatory protein, control, and speed of contraction.

116 Motor control hierarchy

Order the levels of motor control from spinal reflex through brainstem to cortex and state what each level contributes to a voluntary movement.

117 Corticospinal pathway

Trace the corticospinal pathway from the primary motor cortex to the skeletal muscle and identify where it crosses.

118 Cerebellum and basal ganglia

Contrast the contributions of the cerebellum and the basal ganglia to movement and match a described motor sign to the structure involved.

119 Upper and lower motor neuron signs

Distinguish upper from lower motor neuron lesions by tone, reflexes, and muscle bulk and assign a set of findings to the correct level.

Week 8 Hormones and reproduction, the slow control system

Mon Oct 26 to Sun Nov 1 · 25 competencies

120 Signal types and range

Classify a chemical signal as autocrine, paracrine, neurotransmitter, neurohormone, or hormone by its route and distance of travel.

121 Receptor location and ligand solubility

Predict whether a signal molecule binds a surface receptor or an intracellular receptor from its lipid solubility and relate that to the speed and duration of the response.

122 G protein coupled receptors

Trace a G protein coupled receptor pathway from ligand binding through the G protein and amplifier enzyme to the second messenger and the cellular response.

123 Second messengers

Identify cyclic AMP, IP3, diacylglycerol, and calcium as second messengers and state the enzyme that generates each and one target it activates.

124 Catalytic receptors and intracellular receptors

Compare a receptor enzyme such as the insulin receptor with an intracellular steroid receptor by mechanism and by time course of the response.

125 Signal amplification

Explain how a cascade amplifies a signal and estimate the size of the amplification across the steps of a given pathway.

126 Receptor modulation

Define agonist, antagonist, competitive inhibition, up regulation, and down regulation and predict the effect of chronic ligand excess on target cell sensitivity.

127 Signal termination

Name three mechanisms that end a chemical signal and explain why a signal that cannot be terminated produces pathology.

128 Dose response relationships

Plot a dose response curve, identify threshold, maximal response, and EC50, and compare a full agonist with a partial agonist on the same axes.

129 Hormone classes

Classify hormones as peptide, steroid, or amine and compare the three by synthesis, storage, and receptor location.

130 Hormone transport and half life

Explain how protein binding in plasma affects the free hormone fraction and the half life and predict which hormone class circulates bound.

131 Hormone receptors and target response

Explain why one hormone can produce different responses in different tissues and relate receptor number to target cell sensitivity.

132 Hormone interactions

Define synergism, permissiveness, and antagonism and identify each in a described hormone pair.

133 Control of hormone release

Classify a hormone stimulus as humoral, neural, or hormonal and trace an example of each.

134 Hypothalamic pituitary axes

Trace a three tier axis from hypothalamic releasing hormone through the anterior pituitary trophic hormone to the peripheral gland and place the long and short loop negative feedback.

135 Primary and secondary endocrine disorders

Use hormone levels at two tiers of an axis to classify a disorder as primary or secondary and as hypersecretion or hyposecretion.

136 Posterior pituitary hormones

Explain why the posterior pituitary is neural tissue and state the stimulus and target action of antidiuretic hormone and oxytocin.

137 Hormone assay

Run or simulate an immunoassay for a hormone and use the standard curve to determine an unknown concentration.

138 Hypothalamic pituitary gonadal axis

Trace the gonadotropin releasing hormone axis to the gonads and place the feedback loops for both sexes.

139 Male reproductive physiology

Explain the control of spermatogenesis by follicle stimulating hormone, luteinizing hormone, and testosterone and state the roles of the Sertoli and Leydig cells.

140 Ovarian cycle

Order the follicular, ovulatory, and luteal phases and explain how the luteinizing hormone surge is triggered by positive feedback.

141 Uterine cycle

Correlate the menstrual, proliferative, and secretory phases of the uterine cycle with the estrogen and progesterone levels that drive them.

142 Pregnancy and placental hormones

Explain how human chorionic gonadotropin rescues the corpus luteum and state the roles of placental estrogen, progesterone, and relaxin.

143 Parturition and lactation

Explain labor as a positive feedback loop driven by oxytocin and compare the hormonal control of milk production with milk ejection.

144 Hormone cycle graph interpretation

Read a graph of gonadotropin and ovarian hormone levels across a cycle and identify the day of ovulation and the event driving each peak.

Week 9 The heart as a pump

Mon Nov 2 to Sun Nov 8 · 17 competencies

145 Adrenal medulla

Explain why the adrenal medulla is a modified sympathetic ganglion and compare the duration of its circulating catecholamine effect with direct sympathetic innervation.

146 Pacemaker potential

Explain the unstable pacemaker potential of the sinoatrial node by the funny current, calcium entry, and potassium exit and state what sets intrinsic heart rate.

147 Contractile cell action potential

Diagram the ventricular action potential and identify the ion movement responsible for the rapid upstroke, the plateau, and repolarization.

148 Conduction system

Trace an impulse through the sinoatrial node, internodal pathways, atrioventricular node, bundle of His, bundle branches, and Purkinje fibers and explain the purpose of the atrioventricular delay.

149 Cardiac refractory period

Explain why the long refractory period of cardiac muscle prevents tetanus and why that matters for pump function.

150 ECG waves and intervals

Label the P wave, QRS complex, T wave, PR interval, and QT interval and state the electrical event each represents.

151 ECG interpretation

Calculate heart rate from an ECG strip and identify sinus rhythm, tachycardia, bradycardia, and a first degree or complete heart block.

152 ECG recording

Record a lead II ECG, measure the intervals, and correlate each wave with the mechanical event that follows it.

153 Cardiac cycle

Order the phases of the cardiac cycle and state the pressure relationship that opens and closes the atrioventricular and semilunar valves in each phase.

154 Pressure volume loop

Label the four phases of a ventricular pressure volume loop and predict how the loop changes with increased preload, increased afterload, or increased contractility.

155 Heart sounds

Explain the origin of the first and second heart sounds and relate a murmur to valve stenosis or regurgitation by its timing.

156 Stroke volume and ejection fraction

Calculate stroke volume and ejection fraction from end diastolic and end systolic volumes and interpret a reduced ejection fraction.

157 Cardiac output

Calculate cardiac output and cardiac reserve and predict the effect of a change in heart rate or stroke volume on each.

158 Frank Starling relationship

State the Frank Starling law and explain how venous return and end diastolic volume determine the force of the next contraction.

159 Preload afterload and contractility

Distinguish preload, afterload, and contractility and predict how a change in each alters stroke volume.

160 Autonomic regulation of the heart

Explain how sympathetic and parasympathetic input alter heart rate and contractility at the receptor and second messenger level.

161 Heart sound and pulse correlation

Auscultate the heart sounds, locate the valve areas, and correlate the sounds with the pulse and with the cardiac cycle phases.

Week 10 Pressure, flow, and holding blood pressure steady

Mon Nov 9 to Sun Nov 15 · 13 competencies

162 Adrenal cortex

Match the three cortical zones to aldosterone, cortisol, and the adrenal androgens and state the primary action of each.

163 Pressure flow and resistance

Apply the relationship among flow, pressure gradient, and resistance and predict the effect of a change in vessel radius, blood viscosity, or vessel length on flow.

164 Vessel structure and function

Match arteries, arterioles, capillaries, venules, and veins to their role in pressure buffering, resistance, exchange, and capacitance.

165 Arterial blood pressure

Define systolic, diastolic, pulse, and mean arterial pressure, calculate mean arterial pressure, and explain how it is determined by cardiac output and total peripheral resistance.

166 Local control of blood flow

Explain myogenic, metabolic, and endothelial control of arteriolar tone and predict the local flow response to increased tissue metabolism.

167 Capillary exchange and Starling forces

Identify the four Starling forces at a capillary and determine the net direction of fluid movement at the arterial and venous ends.

168 Edema

Explain how increased capillary hydrostatic pressure, reduced plasma protein, increased permeability, or lymphatic blockage each produce edema and match a clinical scenario to its mechanism.

169 Venous return

Explain the skeletal muscle pump, the respiratory pump, and venoconstriction and predict the effect of prolonged standing on venous return and stroke volume.

170 Blood pressure measurement

Measure blood pressure by auscultation, explain the origin of the Korotkoff sounds, and record the response to a postural or exercise challenge.

171 Baroreceptor reflex

Trace the baroreceptor reflex from the carotid sinus and aortic arch through the medulla to the heart and vessels and state the response to a fall in blood pressure.

172 Hormonal control of blood pressure

Describe the roles of angiotensin II, aldosterone, antidiuretic hormone, and atrial natriuretic peptide in long term blood pressure control.

173 Cardiovascular response to exercise

Predict the change in heart rate, stroke volume, cardiac output, total peripheral resistance, and regional blood flow distribution during moderate exercise.

174 Compensation in hemorrhage and shock

Sequence the compensatory responses to blood loss and explain the point at which compensation fails.

Weeks 11 to 15

94 competencies. Midterm 3. Window Monday December 14 at 8 am to Wednesday December 16 at 10 pm.

Week 11 Blood and how the body defends itself

Mon Nov 16 to Sun Nov 22 · 19 competencies

175 Plasma and blood functions

State the three functional categories of blood, transport, regulation, and protection, and name the major plasma proteins and the function of each.

176 Hematocrit and formed elements

Define hematocrit and interpret a value as normal or abnormal in the context of anemia and dehydration.

177 Erythrocyte structure and hemoglobin

Relate the biconcave shape and absent nucleus of the erythrocyte to its function and describe how the four heme groups of hemoglobin carry oxygen.

178 Erythropoiesis and its regulation

Trace the erythropoietin negative feedback loop from tissue hypoxia to increased red cell production and state the nutrients required.

179 Red cell destruction and bilirubin

Trace the fate of hemoglobin after red cell breakdown and explain how failure to clear bilirubin produces jaundice.

180 Leukocytes

Match neutrophils, lymphocytes, monocytes, eosinophils, and basophils to their defensive function and to the condition that raises each.

181 Hemostasis

Order the three phases of hemostasis, vascular spasm, platelet plug formation, and coagulation, and identify the role of von Willebrand factor and thromboxane.

182 Coagulation cascade

Compare the intrinsic and extrinsic pathways to the common pathway and identify the roles of thrombin, fibrin, calcium, and vitamin K.

183 Clot limitation and fibrinolysis

Explain how anticoagulants and plasmin limit and remove a clot and predict the effect of a drug that blocks vitamin K or platelet aggregation.

184 ABO and Rh blood types

Predict agglutination for any donor and recipient ABO and Rh combination and explain the mechanism of hemolytic disease of the newborn.

185 Hematocrit and hemoglobin determination

Determine hematocrit and hemoglobin from a sample or simulation and evaluate the result against reference ranges.

186 Blood typing

Perform simulated ABO and Rh typing and determine compatible donor types for the result obtained.

187 Lymphatic return

State the volume of fluid the lymphatic system returns each day and trace lymph from an interstitial space to the subclavian vein.

188 Innate defenses

Describe the physical, chemical, and cellular innate defenses and explain what makes them nonspecific and immediate.

189 Inflammation and fever

Order the events of the inflammatory response and explain how the cardinal signs arise and why fever aids defense.

190 Adaptive immunity

Compare humoral and cell mediated immunity by the cell responsible, the target, and the mechanism of elimination.

191 Antibody structure and function

Relate antibody structure to antigen binding and name the mechanisms by which antibodies neutralize or mark a target.

192 Immune memory and immunization

Compare the primary and secondary antibody responses and explain how active and passive immunization produce protection.

193 Immune dysfunction

Classify allergy, autoimmunity, and immunodeficiency by what the immune system is doing wrong in each.

Week 12 Digestion, and how you use food for fuel

Mon Nov 23 to Sun Nov 29, Thanksgiving week · 23 competencies

194 Growth hormone

Describe the direct and insulin like growth factor mediated actions of growth hormone and predict the result of excess or deficiency before and after epiphyseal closure.

195 Thyroid hormone

Trace thyroid hormone synthesis, state the metabolic actions of T3 and T4, and match hyperthyroid and hypothyroid signs to the underlying rate of metabolism.

196 Cortisol and the stress response

Describe the metabolic, immune, and cardiovascular actions of cortisol and explain the consequences of chronic elevation.

197 Pancreatic islet hormones

Compare insulin and glucagon by stimulus and by effect on liver glycogen and on plasma glucose and identify the islet cell that secretes each.

198 Diabetes mellitus

Distinguish type 1 from type 2 diabetes by mechanism and explain how untreated hyperglycemia produces polyuria and ketoacidosis.

199 Glucose tolerance testing

Plot a glucose tolerance curve from measured or simulated data and distinguish a normal response from an impaired one.

200 Four digestive processes

Distinguish motility, secretion, digestion, and absorption and identify where in the tract each dominates.

201 Motility patterns

Compare peristalsis, segmentation, and the migrating motor complex by pattern and purpose.

202 Neural and hormonal regulation

Explain short and long reflex control by the enteric nervous system and match gastrin, secretin, cholecystokinin, and glucose dependent insulinotropic peptide to their stimulus and action.

203 Phases of digestive control

Describe the cephalic, gastric, and intestinal phases and state the dominant control mechanism in each.

204 Gastric secretion and mucosal protection

Trace the production of hydrochloric acid by the parietal cell and explain how the mucosal barrier protects the stomach and how ulcers develop when it fails.

205 Pancreatic and biliary secretion

State the components of pancreatic juice and bile and explain how bicarbonate neutralizes chyme and how bile salts emulsify fat.

206 Carbohydrate and protein digestion and absorption

Trace a starch and a protein from mouth to bloodstream naming the enzymes at each step and the transport mechanism at the brush border.

207 Lipid digestion and absorption

Trace a triglyceride through emulsification, lipase action, micelle formation, and chylomicron packaging to the lymphatic route and explain why fat takes a different path than glucose.

208 Liver function and enterohepatic circulation

State the metabolic, storage, and detoxification functions of the liver and trace the enterohepatic circulation of bile salts.

209 Large intestine and microbiome

Describe water and electrolyte absorption in the colon, the contribution of the microbiome, and the defecation reflex.

210 Digestive enzyme experiment

Test the effect of pH, temperature, and substrate on a digestive enzyme and relate the results to the region of the tract where that enzyme works.

211 Absorptive state

Describe the fate of glucose, amino acids, and fats in the absorptive state and identify insulin as the dominant hormone.

212 Postabsorptive state

Explain how glycogenolysis, gluconeogenesis, lipolysis, and ketogenesis maintain plasma glucose during fasting and name the hormones that drive each.

213 Integrated glucose regulation

Predict the hormonal and metabolic response to a carbohydrate meal and to a prolonged fast and explain how plasma glucose stays within range in both.

214 Energy balance and metabolic rate

Define basal metabolic rate, state the factors that raise and lower it, and explain how indirect calorimetry estimates energy expenditure.

215 Thermoregulation

Trace the negative feedback response to heat and to cold through the hypothalamus and the effectors and explain the mechanism of fever.

216 Metabolic rate measurement

Estimate metabolic rate from measured or simulated oxygen consumption and compare resting values with values after activity.

Week 13 Breathing, gas transport, and the fast pH lever

Mon Nov 30 to Sun Dec 6 · 24 competencies

217 Functions and functional zones

State the functions of the respiratory system and distinguish the conducting zone from the respiratory zone by structure and by role in gas exchange.

218 Pressure gradients and airflow

Apply Boyle's law to inspiration and expiration and relate alveolar pressure to atmospheric pressure at each point in a quiet breath.

219 Intrapleural pressure

Explain why intrapleural pressure is subatmospheric and predict what happens to the lung and chest wall when the pleural seal is broken.

220 Compliance and elastic recoil

Define lung compliance and elastic recoil and predict the effect of fibrosis and of emphysema on each.

221 Surfactant and surface tension

Explain how surfactant lowers surface tension and why its absence causes alveolar collapse in the premature newborn.

222 Airway resistance

Identify the main determinant of airway resistance and predict the effect of bronchoconstriction, mucus, and sympathetic stimulation on airflow.

223 Lung volumes and capacities

Define tidal volume, inspiratory and expiratory reserve volume, residual volume, vital capacity, and total lung capacity and read each from a spirogram.

224 Obstructive and restrictive patterns

Use the ratio of forced expiratory volume in one second to forced vital capacity to classify a spirometry result as obstructive or restrictive.

225 Alveolar ventilation and dead space

Calculate minute ventilation and alveolar ventilation and explain why slow deep breathing ventilates the alveoli better than rapid shallow breathing.

226 Spirometry

Record a spirogram, calculate the lung volumes and capacities available from it, and compare the results with predicted values.

227 Partial pressures

Apply Dalton's law to calculate the partial pressure of oxygen in inspired and alveolar air and state the normal partial pressures in alveolar air, arterial blood, and venous blood.

228 Diffusion at the respiratory membrane

Apply the determinants of diffusion to the respiratory membrane and predict the effect of edema, fibrosis, and emphysema on gas transfer.

229 Ventilation perfusion matching

Explain how local hypoxic vasoconstriction and bronchiolar responses match ventilation to perfusion and predict the consequence of a mismatch.

230 Oxygen transport and the oxyhemoglobin curve

Explain the sigmoid shape of the oxyhemoglobin dissociation curve and state the physiological advantage of the steep and the flat portions.

231 Curve shifts

Predict the direction of the oxyhemoglobin curve shift produced by a change in pH, carbon dioxide, temperature, or 2,3 bisphosphoglycerate and state what the shift means for tissue oxygen delivery.

232 Carbon dioxide transport

Name the three forms in which carbon dioxide is carried, state the approximate percentage of each, and trace the chloride shift in a systemic capillary.

233 Bohr and Haldane effects

Distinguish the Bohr effect from the Haldane effect and explain how the two work together at the tissue and at the lung.

234 Oxygen content versus partial pressure

Distinguish oxygen content from oxygen partial pressure and saturation and explain why anemia and carbon monoxide poisoning reduce delivery differently.

235 Respiratory centers

Identify the medullary and pontine respiratory centers and explain how the basic breathing rhythm is generated and modified.

236 Chemoreceptor control

Compare central and peripheral chemoreceptors by stimulus and explain why carbon dioxide is the primary minute to minute drive to breathe.

237 Ventilation in exercise and at altitude

Predict the ventilatory response to exercise and to acute and chronic altitude exposure and explain the acclimatization changes.

238 Ventilatory response testing

Measure the change in breathing rate and depth after hyperventilation, breath holding, and rebreathing and explain each result by the change in carbon dioxide.

239 Buffer systems

Compare the bicarbonate, phosphate, and protein buffer systems by location and speed and explain why bicarbonate is the dominant extracellular buffer.

240 Respiratory control of pH

Explain how a change in ventilation shifts the carbon dioxide and bicarbonate equilibrium and therefore plasma pH.

Week 14 The kidney and body fluid balance

Mon Dec 7 to Sun Dec 13 · 21 competencies

241 Calcium homeostasis

Trace the response to falling plasma calcium through parathyroid hormone, calcitriol, and the actions on bone, kidney, and intestine, and state the role of calcitonin.

242 Kidney functions

List the homeostatic functions of the kidney including water and electrolyte balance, acid base regulation, waste excretion, blood pressure control, and hormone production.

243 Nephron structure and function

Match each nephron segment to its dominant transport function and distinguish a cortical from a juxtamedullary nephron by capability.

244 Three renal processes

Define filtration, reabsorption, secretion, and excretion and write the equation that relates them for any solute.

245 Filtration membrane and selectivity

Describe the three layers of the filtration membrane and predict which plasma components appear in the filtrate based on size and charge.

246 Net filtration pressure and GFR

Calculate net filtration pressure from glomerular hydrostatic pressure, capsular pressure, and colloid osmotic pressure and predict the effect of each change on glomerular filtration rate.

247 Regulation of GFR

Compare myogenic autoregulation, tubuloglomerular feedback, sympathetic control, and hormonal control and predict the effect of afferent and efferent arteriolar constriction on filtration rate.

248 Renal clearance

Calculate the clearance of a substance and use inulin and creatinine clearance to estimate glomerular filtration rate.

249 Clearance and handling inference

Use a clearance value relative to glomerular filtration rate to determine whether a substance is net reabsorbed or net secreted.

250 Proximal tubule reabsorption

Explain how the sodium gradient drives reabsorption of glucose, amino acids, and water in the proximal tubule and state why reabsorption there is described as isosmotic.

251 Transport maximum and renal threshold

Distinguish transport maximum from renal threshold and explain the appearance of glucose in urine when plasma glucose is elevated.

252 Tubular secretion

State the purpose of tubular secretion and name substances secreted including hydrogen ion, potassium, and organic anions such as drugs.

253 Countercurrent mechanism

Explain how the countercurrent multiplier of the loop of Henle and the countercurrent exchanger of the vasa recta build and preserve the medullary osmotic gradient.

254 Antidiuretic hormone and urine concentration

Trace the response to dehydration from osmoreceptor firing through antidiuretic hormone release to aquaporin insertion and concentrated urine.

255 Renin angiotensin aldosterone system

Trace the renin angiotensin aldosterone pathway from the stimulus for renin release to the effects of angiotensin II and aldosterone on vessels, tubules, and blood pressure.

256 Natriuretic peptides

Explain how atrial natriuretic peptide opposes the renin angiotensin aldosterone system and state the stimulus for its release.

257 Potassium handling

Explain how aldosterone and plasma potassium regulate potassium secretion in the collecting duct and predict the effect of a potassium wasting diuretic.

258 Micturition

Trace the micturition reflex and explain how the internal and external sphincters allow voluntary control.

259 Urinalysis

Perform a urinalysis measuring specific gravity, pH, glucose, protein, ketones, and blood and relate any abnormal finding to the renal process that failed.

260 Renal function calculation

Calculate glomerular filtration rate, clearance, and filtered load from a data set and interpret the values in a clinical case.

261 Volume and osmolarity disturbances

Classify a disturbance as volume depletion or overload and as hypoosmotic, isosmotic, or hyperosmotic and predict the resulting fluid shift between compartments.

Week 15 The slow pH lever, and putting it all together

Mon Dec 14 to Wed Dec 16, three days · 7 competencies

262 Renal control of pH

Describe how the kidney secretes hydrogen ion and reclaims or generates bicarbonate and explain why renal compensation is slower and more complete than respiratory compensation.

263 The four acid base disorders

Classify a disorder as respiratory or metabolic and as acidosis or alkalosis from pH, partial pressure of carbon dioxide, and bicarbonate.

264 Compensation

Determine whether an acid base disorder is uncompensated, partially compensated, or fully compensated and name the system doing the compensating.

265 Arterial blood gas interpretation

Interpret a set of arterial blood gas values and match the result to a plausible clinical cause.

266 Multisystem integration case

Given a clinical scenario, trace the disturbance through at least three organ systems and identify the compensatory responses in the order they occur.

267 Exercise as an integrated response

Build an integrated account of moderate exercise across the muscular, cardiovascular, respiratory, endocrine, and renal systems.

268 Drawing based synthesis check

Draw and annotate a homeostatic pathway from memory and explain each step aloud or in writing without notes.

If one of these does not make sense yet

That is what the week is for. A competency is written as the finished skill, the thing you can do on the far side of the week, not the thing you already know walking in. Bring the ones that stay stubborn to my Virtual Office and we will take them apart.

Dr. Sharilyn Rennie